

Andrew Harwood

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EDUCATION

Bachelor of Science in Physics

University of Florida Honors Program, Gainesville, FL

May 2027

GPA: 3.96 / 4.00

Minor in Mathematics

Relevant Coursework: Quantum Mechanics, Statistical Mechanics Thermal Physics, Electromagnetism

RESEARCH EXPERIENCE

Undergraduate Researcher

Biophysics Theory, BingKan Xue Group, University of Florida, Department of Physics

October 2024 – Present

- Modeling Bryozoan autozooids using Voronoi diagrams to describe individual orientations and colonial growth patterns
- Comparing calculated Voronoi diagrams with segment data from DeepBryo (Di Martino et al., 2023) to refine the location of the autozoid center using loss functions
- Applied 2-dimensional Fourier analysis on 8,000+ images of Microporella Bryozoan fossils to determine spatial periodicity
- Determined the growth of Microporella colonies to be modified hexagonal
- Discovered that the autozooidal orifice exhibits stronger hexagonal periodicity than the center of mass of the autozoid
- Presented results to other researchers from multiple groups at a biophysics seminar

TECHNICAL EXPERIENCE

Avionics and Recovery Lead

NASA USLI, Swamp Launch Rocket Team

June 2024 – June 2025

- Organized subteam meetings to provide project updates, assign and oversee tasks, and integrate individual work
- Designed and manufactured a subscale and full-scale launch vehicle avionics bay
- Wired and soldered altimeters, arming switches, ejection charge terminals, and GPS circuits capable of sustaining 9.0+ G
- Programmed and tested two barometric altimeters using a vacuum chamber to simulate ascent and descent
- Performed live-charge ejection tests to ensure the success of proper parachute deployment
- Integrated and operated a TeleGPS position tracking device on the 70 cm band using radio frequency and a Yagi antenna
- Analyzed descent rates and in-flight forces to select and test parachutes, recovery harnesses, and recovery hardware
- Demonstrated technical writing to NASA in five design reviews, which included decision matrices, CAD models and part drawings, electrical schematics, and comprehensive manufacturing documentation
- Placed 6th out of 53 Universities, Altitude Award, AIAA Reusable Launch Vehicle Award, 2nd place STEM engagement

INVOLVEMENT

Member

- Society of Physics Students
- USLI Payload Mechanics, Swamp Launch Rocket Team
- Academy of Model Aeronautics
- South Park Area Radio Kontrol Society

August 2024 – Present

September 2023 – May 2024

Musician

- Clarinet Ensemble, New Music Ensemble, Steel Drum Ensemble, Symphonic Band

SKILLS AND CERTIFICATIONS

Programming: Python, Jupyter Notebook, Pandas, SciPy, Matplotlib, scikit-image, pycocotools, OpenCV, Fusion 360, OpenRocket

Manufacturing: Soldering, 3D printing, carbon-fiber layups, CNC

Languages: English, French (Working Proficiency)

Awards:

President's Honor Roll

Fall 2023, Spring 2024, Spring 2025

Dean's List

Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026

The Johns Hopkins University Book Award

May 2022

Rotary Club Student of the Month

April 2023

PERSONAL PROJECTS

NeoPixel Replica Lightsaber

- Designed and integrated a SaberTec GHv3 soundboard along with a suite of microelectronics into a compact, 3D-printed modular chassis for a machined aluminum hilt
- Utilized rotary pin connectors and NeoPixel LED connectors to allow the replica's soundboard and LED Pixel blade to be hot-swappable and fully programmable